

Pathmika Weerarathna

Associate Tech Lead — Full-Stack Engineer (Mobile & Web), AI/ML Exposure

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Summary

Full-stack engineer and Associate Tech Lead with nearly five years of experience delivering production mobile and web applications for telecom operators, having grown through every engineering level at Omobio. Contributed to 25+ client and product engagements – delivery, architecture, and pre-sales solutioning – across 9+ countries in Asia, Africa, and Latin America. Core strengths in React Native mobile development (native modules, app security, monetization) and Next.js/React web platforms, backed by solid REST API and backend engineering. Alongside this, gained hands-on AI/ML exposure self-initiating and leading a telecom fraud-detection R&D initiative spanning NLP embeddings, GenAI, and ensemble ML models. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 – Present

Omobio (Pvt) Ltd

- Lead, mentor, and scale a cross-functional engineering team of 10+ developers, driving solution design and delivery across mobile, web, and data-platform initiatives.
- Serve as technical subject-matter expert for pre-sales engagements, producing architecture proposals and technical validation.
- Lead two flagship platforms: the AI/ML R&D team enhancing the SMS Firewall product with ML-driven detection, and the SOR (Steering of Roaming) centralized reporting platform.
- Manage client engagements end-to-end.

Senior Software Engineer

Jan 2024 – Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end mobile and web applications, owning architecture design and effort estimation.
- Delivered SQL/CDR reporting engineering for Banglalink's SMSC billing platform (Bangladesh), recovering ~85,000 dropped records and fixing a 240x query-performance regression.
- Architected cross-platform mobile solutions in React Native with deep native integration (Java, Kotlin, Swift), including a ~2,700-line custom native alarm engine built from the ground up across 26 files, replacing third-party alarm libraries on Android and iOS.
- Integrated RevenueCat for end-to-end in-app subscription purchasing.
- Self-initiated AI/ML capabilities in telecom security from the ground up, building a production fraud-detection module combining semantic embeddings and a local-LLM GenAI layer (see Key Projects).
- Built HOPP, Hutch's Next.js 14 web self-care portal — authoring the core auth/locale middleware, a custom Axios API layer, and end-to-end integration across six payment methods.
- Continued ownership of Hutch Selfcare (led for 2+ years across this role and the one prior): led its React Native version upgrade from 0.66.5 to 0.76.5 and delivered 3+ change requests for new feature additions.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards and GitHub Actions CI/CD pipelines to improve software reliability.

Software Engineer

Apr 2022 – Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Contributed to Dialog's enterprise web platforms (MDA, CDR-disclosure portal) using PHP Yii2 and React.
- Hardened Hutch Selfcare against fraud with certificate/public-key pinning and a device-integrity gate, and led the React Native version upgrade of Vodafone Fiji's production self-care app from RN 0.59.3 to 0.70.5.
- Participated in pre-sales client engagement and contributed to ISO 27001 certification-readiness documentation and a company-wide Git branching standard, later delivered as an internal training program.

- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

Software Engineer (Intern) – Full Stack

Oct 2021 – Apr 2022

Omobio (Pvt) Ltd

- Supported frontend, backend, and mobile feature development for enterprise-grade projects using React.js, React Native, Python, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **Hutch Selfcare (Telecom Self-Care App), 2+ years:** Owned and evolved Hutch's flagship React Native app — security hardening, dual FCM/HMS push architecture, a RN version upgrade (0.66.5 → 0.76.5), and 3+ feature change requests.
- **HOPP Web Self-Care Portal, 2024:** Next.js/TypeScript portal with custom Axios API layer and payment-gateway integration, delivered under a Hybrid Git Flow model.
- **SMS Firewall AI/ML Suite:** Lead the AI/ML detection layer for a telecom SMS-firewall product — semantic embeddings, a local-LLM GenAI subsystem, and an XGBoost/CatBoost ensemble classifier for grey-route and OTP-bypass detection.
- **SOR — Centralized Reporting Platform:** Lead a centralized telecom data-reporting platform spanning backend pipelines and web dashboards, with asynchronous task processing and containerized deployment.
- **Circadian — Native Alarm Engine & Monetization, 2024–2025:** Fully custom cross-platform alarm engine (reboot-persistence, background/killed-app handling, astronomy-driven scheduling) with RevenueCat-based subscription and entitlement management.
- **Vodafone Fiji RN Upgrade, 2022:** Led a major React Native version upgrade with New Architecture scaffolding on a large, multi-year production codebase.

Additional Experience & Engagements

Self-initiated the AI/ML fraud-detection R&D work above from the ground up (embedding-based detection, a local-LLM GenAI subsystem, and an XGBoost/CatBoost ensemble classifier), and separately architected FirewallBI, the large-scale telecom BI platform underpinning it (2023–2025; 8 on-prem sites, 300+ Logstash pipelines, mutual-TLS Elasticsearch cluster).

Also contributed to 15+ further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms; Robi's Bangla SMS Opt-In/Opt-Out solution scoping (2023); the RoamerSteering ELK analytics platform (2023); and solution proposals for Ncell (Nepal), Grameenphone (Bangladesh), and Ethio Telecom (Ethiopia).

Areas of Expertise

- **Mobile:** React Native (incl. New Architecture), Android (Java/Kotlin), iOS (Swift/Objective-C), Huawei HMS, native module bridging
- **Web:** React.js, Next.js, TypeScript/Javascript, Redux/Redux-Saga
- **Backend & APIs:** Python, FastAPI, REST API design, Microservices, Java (Spring Boot), .NET (ASP.NET Core)
- **AI/ML (exposure):** PyTorch, Hugging Face Transformers, LangChain, XGBoost, CatBoost, SHAP, NLP, Semantic Search
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Airflow, Celery
- **DevOps:** Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- **Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) – University of Moratuwa

July 2025 – Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 – University of Colombo School of Computing
2019 – 2022

Diploma in English (Certificate Level), Distinction Pass – Seagis Campus, Nugegoda

2018

Achievements

- Emerging Employee of the Year – Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee – University of Colombo (2021): achieved GPA above 3.7 in first academic year.

References available upon request.